Introduction

1. **Transcriptional Dynamics During Mouse Tissue Development**

Mammalian development is governed by tightly regulated, stage-specific transcriptional programs that orchestrate cell differentiation, tissue maturation, and functional specialization. These programs evolve dynamically over developmental time and diverge between tissues, reflecting distinct cellular identities and physiological roles. Characterizing how gene expression landscapes change across developmental stages and between organs is therefore fundamental for understanding the molecular principles underlying organogenesis and postnatal maturation.

High-throughput RNA sequencing (RNA-seq) has become a central technology for studying transcriptional dynamics at genome scale. RNA-seq enables unbiased quantification of gene expression, supports complex experimental designs, and allows the integration of multivariate exploratory analyses with rigorous statistical modeling. When combined with functional enrichment analyses and genome-level visualization, RNA-seq provides a comprehensive framework for linking transcriptional changes to biological processes and developmental trajectories.

In this project, we analyze a publicly available RNA-seq dataset generated to map transcriptional dynamics during mouse development, focusing exclusively on the protein-coding mRNA transcriptome. The dataset comprises paired-end RNA-seq libraries from Mus musculus (strain C57BL/6J, wild-type), sampled across two biologically distinct tissues—brain and liver—and six developmental stages spanning embryonic to postnatal development: embryonic day 15.5 (E15.5), embryonic day 18.5 (E18.5), postnatal day 0.5 (P0.5), postnatal day 4 (P4), postnatal day 22 (P22), and postnatal day 29 (P29). For each tissue–stage combination, two independent biological replicates were generated, resulting in a balanced experimental design well suited for multivariate analysis and differential expression modeling.

The overarching goal of this study is to characterize how transcriptional programs are remodeled during development and to compare developmental expression trajectories between liver and brain. Specifically, we aim to (i) assess global expression patterns across samples, (ii) identify genes that are differentially expressed between key developmental stages while controlling for tissue identity, (iii) interpret these changes in terms of biological processes, and (iv) visually validate representative transcriptional differences at the genome level.

2. **Dataset Description and Experimental Design**

This study analyzes a balanced RNA-sequencing dataset comprising **24 paired-end samples** generated from Mus musculus and aligned to the mm10 reference genome. The dataset is publicly available under accession E-MTAB-2328 and was selected due to its high quality, clear experimental structure, and relevance to developmental transcriptomics.

Samples originate from two biologically distinct tissues, brain and liver, and span six developmental stages covering embryonic and postnatal development (E15.5, E18.5, P0.5, P4, P22, and P29). For each tissue–stage combination, two independent biological replicates were generated, yielding a fully crossed and balanced design. This structure enables the separation of tissue-specific expression effects from developmental stage–dependent changes, while providing sufficient replication for robust statistical inference.

All libraries were prepared using strand-specific, ribosomal RNA–depleted protocols and sequenced in paired-end mode. This experimental setup enables accurate gene-level quantification of protein-coding transcripts and supports downstream analyses of global expression structure, differential expression, and functional enrichment.

3. Computational Analysis Strategy

All analyses were conducted using a fully reproducible RNA-seq pipeline following established best practices. Each processing step was logged, and outputs were organized within a transparent folder structure to ensure traceability and reproducibility.

Raw sequencing reads were initially assessed using quality control metrics to evaluate per-base sequence quality, adapter contamination, and sequence composition. Adapter trimming and quality filtering were subsequently applied to remove low-quality bases and technical artifacts. Post-trimming quality control confirmed substantial improvements in read quality, particularly with respect to adapter content and base-call accuracy.

High-quality trimmed reads were aligned to the mm10 mouse reference genome using the STAR aligner, producing coordinate-sorted BAM files for each sample. Alignment quality was evaluated using standard metrics, including uniquely mapped read percentage, multi-mapping rate, and mismatch frequency. All samples exhibited high mapping rates consistent with high-quality RNA-seq data.

Gene-level quantification was performed using featureCounts, aggregating exon-level reads into gene counts based on annotated gene identifiers. This step yielded a unified gene count matrix across all samples, which served as the input for downstream statistical analyses.

4. Statistical Modeling and Exploratory Data Analysis

Differential gene expression analysis was performed using DESeq2, which models RNA-seq count data using a negative binomial generalized linear model. To estimate developmental stage effects while accounting for tissue-specific expression differences, a design formula of the form:

$$\text{Design} = \sim \text{tissue} + \text{stage}$$

was employed. Interaction terms were intentionally excluded in order to focus on main biological effects and to maintain model interpretability, in line with the primary objectives of the project.

Prior to differential expression testing, normalized counts were transformed using the variance-stabilizing transformation (VST). Principal component analysis (PCA) based on VST-normalized counts was used for quality control and exploratory analysis. The PCA revealed a dominant separation by tissue identity along the first principal component, explaining the majority of variance, and a secondary separation by developmental stage along subsequent components. This structure indicates that biological factors strongly outweigh technical variability in the dataset.

Differential expression analyses were performed for representative developmental contrasts, focusing on transitions from embryonic to postnatal stages. Results were summarized using volcano plots and heatmaps of top differentially expressed genes. Functional interpretation of significantly differentially expressed genes was conducted using Gene Ontology (GO) Biological Process enrichment analysis, providing insight into the cellular and molecular programs associated with developmental progression.

Finally, representative differentially expressed genes were visually validated at the genome level using CPM-normalized bigWig tracks in the Integrative Genomics Viewer (IGV). This step ensured concordance between statistical results and underlying read coverage patterns across biological replicates.

5. **Objectives of the Study**

The analytical framework of this project was designed to address the following core objectives:

1. Assess global transcriptional structure across samples and verify that tissue identity and developmental stage drive the dominant expression patterns.
2. Identify differentially expressed genes associated with key developmental transitions, focusing on representative contrasts (P29 vs E15.5 and P4 vs E15.5).
3. Characterize functional programs underlying developmental changes through Gene Ontology Biological Process enrichment analysis.
4. Visually validate transcriptional changes at the genome level using normalized bigWig tracks and IGV snapshots of top differentially expressed genes.

By integrating statistical modeling, multivariate visualization, functional enrichment, and genome-level signal inspection, this study provides a comprehensive and biologically interpretable analysis of transcriptional dynamics during mouse tissue development.

Results

1. Quality Control and Exploratory Analysis: VST and PCA

To ensure that downstream multivariate analyses capture biologically meaningful variation rather than technical noise, gene expression counts were transformed using the variance stabilizing transformation (VST) implemented in DESeq2. VST reduces the dependence of variance on the mean expression level, thereby preventing highly expressed genes from disproportionately driving the analysis and improving sensitivity to relative expression patterns across samples.

Principal component analysis (PCA) was subsequently performed on the VST-normalized counts to assess global transcriptional structure, sample relationships, and data quality. PCA revealed a clear and biologically coherent organization of samples. The first principal component (PC1), explaining 91% of the total variance, strongly separates samples by tissue identity, distinguishing brain from liver across all developmental stages. This dominant separation indicates that tissue-specific transcriptional programs represent the primary source of variation in the dataset.

The second principal component (PC2), accounting for 5% of the variance, captures variation associated with developmental stage, reflecting progressive transcriptional remodeling during embryonic and postnatal development within each tissue. Importantly, biological replicates cluster tightly within their respective tissue–stage groups, demonstrating high reproducibility and the absence of pronounced batch effects or technical confounders.

Overall, the PCA confirms excellent data quality and validates the experimental design, supporting the use of downstream differential expression and functional enrichment analyses.

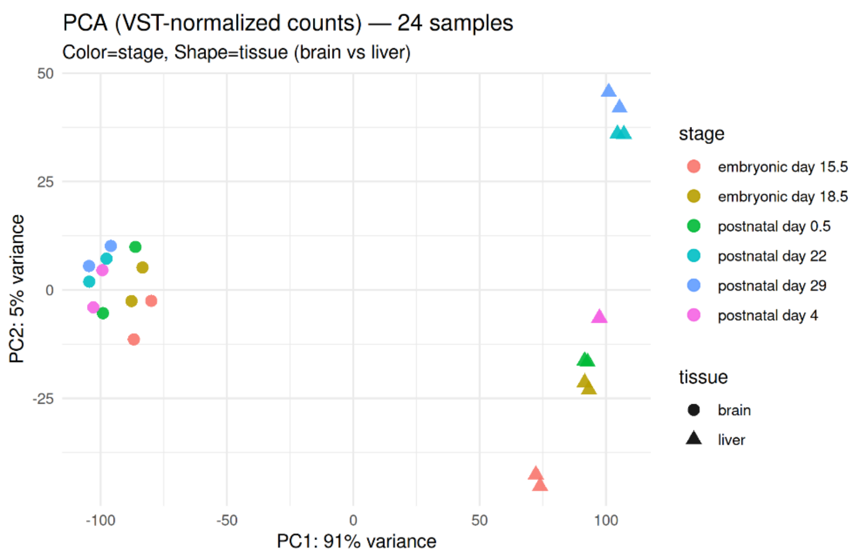

Figure 1 Principal component analysis (PCA) of VST-normalized RNA-seq expression profiles.

PCA was performed on variance-stabilized gene expression counts from 24 RNA-seq samples. Each point represents one biological sample, with color indicating developmental stage and shape indicating tissue (brain or liver). The first principal component (PC1; 91% variance explained) separates samples by tissue identity, highlighting strong tissue-specific transcriptional programs. The second principal component (PC2; 5% variance explained) reflects developmental stage–associated variation within each tissue. Tight clustering of biological replicates within groups indicates high reproducibility and minimal batch effects.

2. Differential Gene Expression Analysis (DESeq2)

Differential gene expression analysis was performed using DESeq2, modeling raw gene counts with a negative binomial generalized linear model. The design formula was specified as:

$$\text{design} = \sim \text{tissue} + \text{stage}$$

This formulation estimates the effect of developmental stage while controlling for tissue identity, effectively removing the systematic expression offset between brain and liver prior to stage-wise comparisons. Interaction terms were intentionally excluded in order to focus on global developmental effects shared across tissues and to maintain a clear and interpretable statistical framework.

Two biologically relevant developmental contrasts were examined:

- Postnatal day 29 (P29) vs embryonic day 15.5 (E15.5)

  A total of 10,915 genes were identified as significantly differentially expressed (adjusted p-value < 0.05), of which 6,004 were upregulated and 4,911 were downregulated at P29 relative to E15.5. This large number of differentially expressed genes reflects extensive transcriptional remodeling between late embryonic and mature postnatal stages.

- Postnatal day 4 (P4) vs embryonic day 15.5 (E15.5)

  In this earlier developmental transition, 3,670 genes were significantly differentially expressed (adjusted p-value < 0.05), including 2,337 upregulated and 1,333 downregulated genes. The smaller number of affected genes is consistent with a more gradual transcriptional shift at early postnatal stages.

Statistical significance was defined using Benjamini–Hochberg false discovery rate (FDR) correction, and genes with both adjusted p-value < 0.05 and absolute $\log_2$ fold change ≥ 1 were considered strongly differentially expressed and highlighted in downstream visualizations.

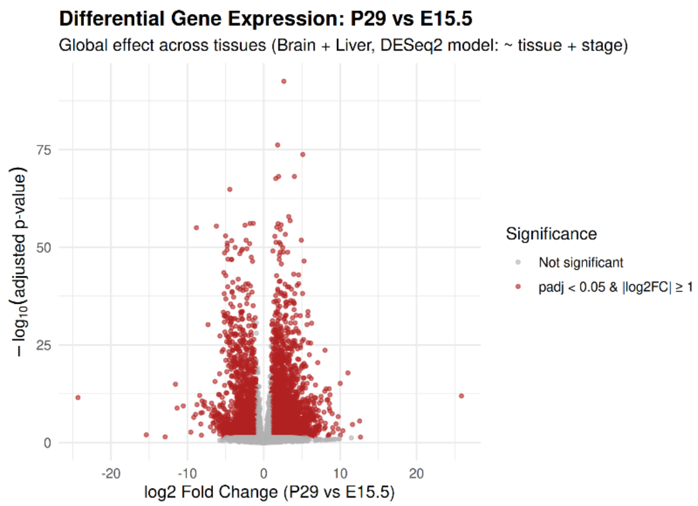

Figure 2 Volcano plot of differential gene expression for P29 vs E15.5.

The volcano plot displays genome-wide differential expression results comparing postnatal day 29 (P29) to embryonic day 15.5 (E15.5), based on a DESeq2 model accounting for tissue effects (design: ~ tissue + stage). Each point represents one gene, plotted by $\log_2$ fold change (x-axis) and $-\log_{10}$ adjusted p-value (y-axis). Genes meeting the significance criteria (adjusted p-value < 0.05 and $|\log_2 FC| \geq 1$) are highlighted in red, while non-significant genes are shown in grey. The widespread distribution of significant genes across both positive and negative fold changes reflects extensive transcriptional reprogramming associated with late developmental maturation across tissues.

3. Expression Patterns of Top Differentially Expressed Genes

Hierarchical clustering of the top 20 differentially expressed genes identified in the P29 versus E15.5 contrast reveals clear and biologically coherent expression patterns across samples. Genes segregate into distinct clusters corresponding to developmental stage, with late postnatal samples (P22–P29) showing coordinated upregulation or downregulation relative to embryonic stages. Sample clustering further reflects strong tissue-specific structure, while maintaining consistent replicate grouping within each tissue–stage combination. These patterns confirm the robustness of the differential expression results and highlight gene subsets that undergo pronounced transcriptional reprogramming during late developmental maturation.

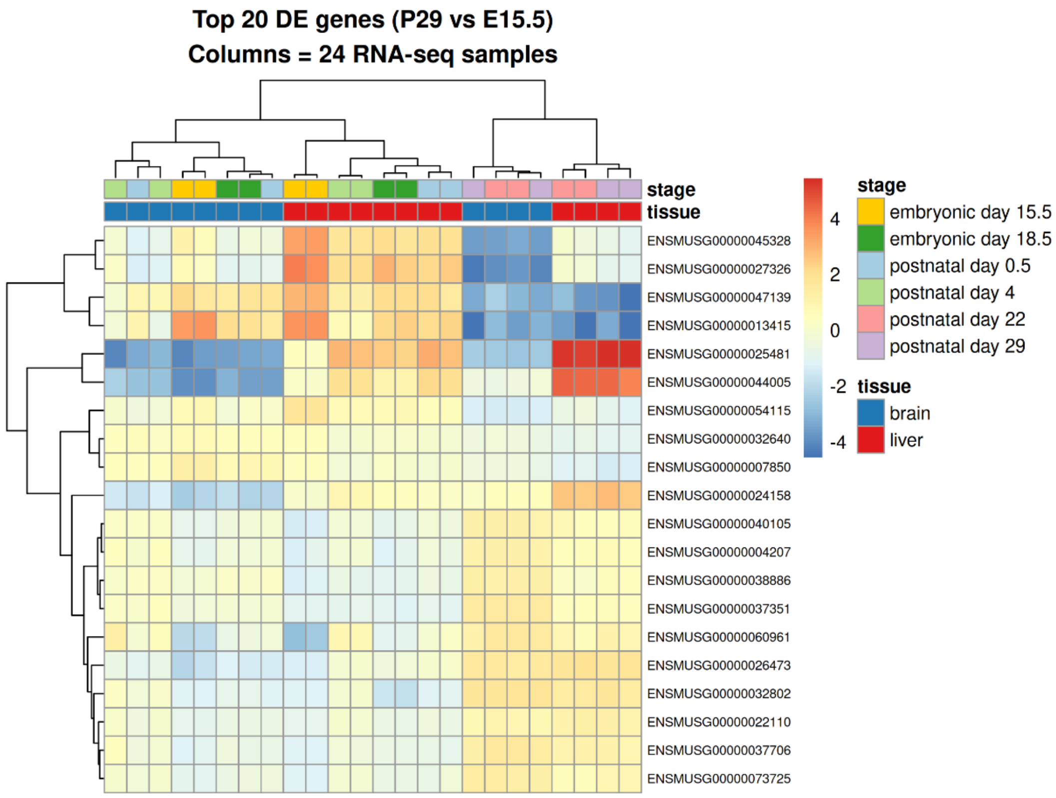

Figure 3 Heatmap of the top 20 differentially expressed genes in the P29 vs E15.5 contrast.

The heatmap displays variance-stabilized expression values (VST) for the top 20 genes ranked by statistical significance in the P29 versus E15.5 comparison, across all 24 RNA-seq samples. Rows correspond to genes and columns to individual samples. Both genes and samples were hierarchically clustered based on expression similarity. Sample annotations indicate developmental stage (color-coded) and tissue identity (brain vs liver). The color scale represents relative expression levels, with warmer colors indicating higher expression and cooler colors indicating lower expression. The observed clustering highlights coherent stage-dependent transcriptional programs and consistent replicate behavior across tissues.

4. Differential Expression Summary for the P29 vs E15.5 Contrast

The table summarizes representative differentially expressed genes identified in the P29 versus E15.5 comparison using a DESeq2 model that adjusts for tissue effects. Genes with positive log2 fold change values are more highly expressed at postnatal day 29, whereas negative values indicate higher expression at embryonic day 15.5, after controlling for systematic brain–liver differences. The extremely low adjusted p-values across entries indicate strong statistical support for these expression changes, reflecting robust developmental regulation rather than tissue-driven effects. Together, these results highlight extensive transcriptional remodeling between embryonic and late postnatal stages.

baseMean	log2FoldChange	padj
3308,379911	2,633819606	3,25804E-93
3708,577036	1,818532275	6,70389E-77
870,8092642	5,116887182	1,80172E-74
2925,223909	4,012713211	7,53757E-69
501,8347359	1,94485497	7,53757E-69
2173,161526	1,577589688	2,60541E-68
2028,120993	-4,438965857	1,51082E-65
26505,60005	3,268743681	1,42697E-58
4452,914135	3,456383134	1,62556E-57
1398,411761	-1,382299758	7,59995E-57
9127,740726	-1,802309014	8,2724E-57
16649,90409	1,876840826	9,20277E-57
6232,001921	2,281712806	1,86412E-56
630,6084255	-2,456892983	2,33006E-56
8951,187858	-6,193529093	3,90055E-56
3317,551153	1,70284719	7,32507E-56
483,3321019	-8,809157136	9,84564E-56
2824,411648	2,16110781	2,79913E-55
1286,770292	2,827364213	5,12497E-54
2888,503064	-5,001138395	1,22709E-53

Table 1 Differential expression statistics for selected genes in the P29 vs E15.5 comparison.

The table reports DESeq2 results for representative genes, including mean normalized expression (baseMean), log2 fold change (log2FoldChange), and Benjamini–Hochberg adjusted p-values (padj). Positive log2FoldChange values indicate higher expression at postnatal day 29 relative to embryonic day 15.5, while negative values indicate higher expression at E15.5. Differential expression was estimated using a model that accounts for tissue effects (~ tissue + stage), ensuring that reported changes reflect developmental stage differences rather than tissue-specific expression biases.

5. Gene Ontology (Biological Process) Enrichment Analysis

Gene Ontology (GO) Biological Process enrichment analysis was performed on genes significantly differentially expressed in the P29 versus E15.5 comparison (adjusted p-value < 0.05). The enrichment analysis is based on genes identified from the global developmental stage effect estimated by the DESeq2 model while controlling for tissue-specific expression differences. The top enriched terms ranked by adjusted p-value are dominated by processes related to cell cycle progression and mitotic regulation, including chromosome segregation, DNA replication, and nuclear division. In parallel, multiple metabolic processes—such as fatty acid metabolism, oxidative phosphorylation, and cellular respiration—are significantly overrepresented. This combined enrichment pattern reflects a major developmental transition from embryonic proliferative programs at E15.5 toward enhanced metabolic specialization and tissue maturation at the late postnatal stage (P29).

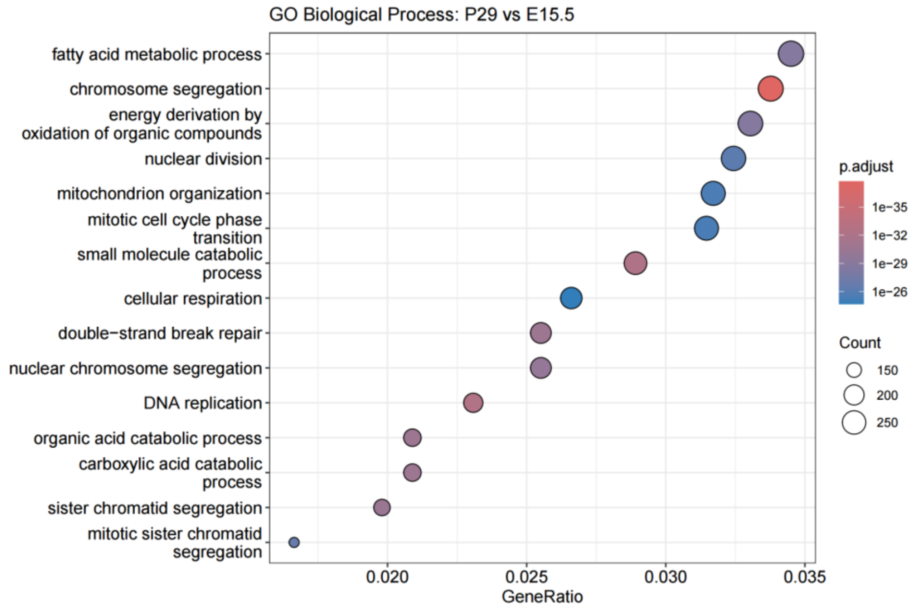

Figure 4 Gene Ontology (GO) Biological Process enrichment for differentially expressed genes between P29 and E15.5.

Dot plot showing the top enriched GO Biological Process terms for genes significantly differentially expressed in the P29 versus E15.5 contrast (adjusted p-value < 0.05), based on a DESeq2 model accounting for tissue effects (~ tissue + stage). The x-axis represents the GeneRatio (proportion of significant genes associated with each term), dot size indicates the number of contributing genes (Count), and color encodes the Benjamini–Hochberg adjusted p-value. Enriched terms are dominated by cell cycle– and mitosis-related processes as well as metabolic and mitochondrial pathways, reflecting a coordinated transition from embryonic proliferative programs to postnatal metabolic maturation.

Description	GeneRatio	p.adjust	Count
chromosome segregation	278/8232	1,86E-38	278
DNA replication	190/8232	3,16E-33	190
small molecule catabolic process	238/8232	5,26E-33	238
double-strand break repair	210/8232	2,22E-31	210
organic acid catabolic process	172/8232	2,22E-31	172
carboxylic acid catabolic process	172/8232	2,22E-31	172
sister chromatid segregation	163/8232	3,19E-31	163
nuclear chromosome segregation	210/8232	1,16E-30	210
fatty acid metabolic process	284/8232	1,93E-29	284
energy derivation by oxidation of organic compounds	272/8232	1,93E-29	272
mitotic sister chromatid segregation	137/8232	1,62E-27	137
nuclear division	267/8232	6,16E-27	267
mitochondrion organization	261/8232	3,13E-26	261
mitotic cell cycle phase transition	259/8232	3,75E-26	259
cellular respiration	219/8232	2,1E-25	219
mitotic nuclear division	178/8232	9,42E-25	178
lipid transport	261/8232	3,5E-24	261
lipid localization	271/8232	7,23E-24	271
sulfur compound metabolic process	186/8232	7,87E-24	186
macroautophagy	212/8232	7,87E-24	212
nucleoside phosphate biosynthetic process	198/8232	1,21E-23	198
organic anion transport	270/8232	2,37E-23	270
DNA recombination	205/8232	7,09E-23	205
purine nucleotide metabolic process	260/8232	1,32E-22	260
establishment of protein localization to organelle	259/8232	1,43E-22	259
vacuole organization	170/8232	1,43E-22	170
aerobic respiration	191/8232	2,89E-22	191
oxidative phosphorylation	107/8232	7,69E-22	107
regulation of lipid metabolic process	225/8232	9,25E-22	225
negative regulation of cell cycle	220/8232	1,09E-21	220
regulation of cell cycle phase transition	249/8232	3,69E-21	249

Table 2 Top enriched GO Biological Process terms for the P29 versus E15.5 comparison.

The table reports GeneRatio, number of contributing genes (Count), and Benjamini–Hochberg adjusted p-values for the most significantly enriched Biological Process terms identified from differentially expressed genes.

6. IGV-based visual validation of differential expression

To visually validate the differential expression results, CPM-normalized bigWig coverage tracks were generated from the aligned BAM files and inspected in IGV using the mm10 reference genome. For each tissue, the top five differentially expressed (DE) genes were selected based on adjusted p-value ranking (padj), and genome browser snapshots were obtained. The representative example shown here corresponds to the top-ranked DE gene (rank 1) in the P29 versus E15.5 contrast. For liver and brain separately, two biological replicates per developmental stage are displayed. In both tissues, the coverage profiles show high concordance between replicates within the same stage and a clear, stage-dependent difference in signal intensity between E15.5 and P29. This qualitative agreement between normalized read coverage and DESeq2-derived statistics provides independent, genome-level confirmation of the robustness and biological relevance of the differential expression findings.

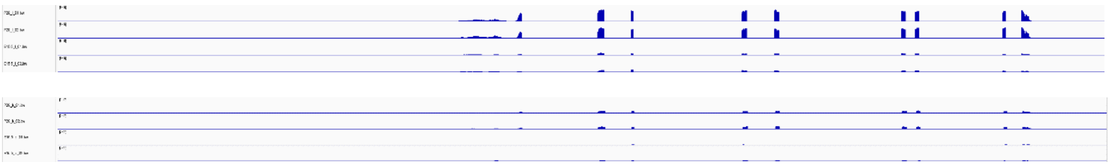

Figure 5 IGV visualization of the top differentially expressed gene between P29 and E15.5 (up liver, down brain).

CPM-normalized bigWig coverage tracks are shown for the highest-ranked differentially expressed gene identified by DESeq2 in the P29 versus E15.5 contrast. The upper panel displays liver samples and the lower panel brain samples, with two biological replicates per developmental stage (E15.5 and P29). Blue coverage profiles represent read depth across the gene locus. Replicates within each stage exhibit highly consistent signal patterns, while a clear stage-dependent difference in coverage is observed, providing genome-level visual validation of the differential expression detected by DESeq2.

Conclusion

In this project, we performed a comprehensive RNA-seq analysis to characterize transcriptional dynamics during mouse development across two biologically distinct tissues, brain and liver. Using a balanced experimental design spanning embryonic to postnatal stages, we applied a reproducible computational pipeline combining rigorous quality control, normalization, multivariate analysis, differential expression testing, functional enrichment, and genome-level visualization.

Exploratory analyses based on VST-normalized counts revealed that tissue identity is the dominant source of transcriptional variation, with developmental stage contributing a secondary but biologically meaningful effect. Differential expression analyses identified thousands of genes whose expression changes significantly between early embryonic (E15.5) and postnatal stages (P4 and P29), even after controlling for tissue-specific differences. Functional enrichment highlighted coordinated regulation of cell cycle–related processes and metabolic pathways, reflecting the transition from proliferative embryonic programs to metabolically mature postnatal states.

Importantly, statistical findings were independently supported by IGV-based visualization of normalized read coverage, demonstrating consistent replicate behavior and clear stage-dependent expression differences at the gene level. Together, these results provide a coherent and biologically interpretable view of developmental transcriptional remodeling in mouse tissues. The integrated analytical framework presented here illustrates how RNA-seq data can be leveraged to link global expression patterns, gene-level changes, functional pathways, and raw genomic signal, offering a robust foundation for studying developmental gene regulation.

References

[1] Schmitt, B. M., et al. High-resolution mapping of transcriptional dynamics across tissue development reveals a stable mRNA–tRNA interface. *Genome Research*, 24(11), 1797–1807 (2014).

[2] Love, M. I., Huber, W., & Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. * Genome Biology*, 15(12), 550 (2014).

[3] Dobin, A., et al. STAR: ultrafast universal RNA-seq aligner. *Bioinformatics*, 29(1), 15–21 (2013).

[4] Yu, G., et al. clusterProfiler: an R package for comparing biological themes among gene clusters. *OMICS: A Journal of Integrative Biology*, 16(5), 284–287 (2012).

Deliverables and Accompanying Files

The following outputs were generated and are provided as part of the main report and supplementary material:

- **Quality control and exploratory analysis**
  - Principal Component Analysis (PCA) based on variance-stabilized (VST) counts, used for quality assessment and evaluation of sample clustering by tissue and developmental stage.
- **Differential expression analysis: P29 vs E15.5 (global across tissues)**
  - Volcano plot highlighting significantly differentially expressed genes.
  - Heatmap of the top differentially expressed genes across all samples.
  - Top differentially expressed genes table (ranked by adjusted p-value).
  - Complete differential expression results table (full gene list; Excel/format).

- **Differential expression analysis: P4 vs E15.5 (global across tissues)**
    - Complete differential expression results table.
    - Top differentially expressed genes table.
- **Genome-level visual validation**
    - IGV snapshots (CPM-normalized bigWig tracks) for the top five differentially expressed genes per tissue (brain and liver), including two biological replicates per stage.
- **Functional enrichment analysis**
    - Enrichment plots (barplot/dotplot) and a table of top enriched GO terms.

All figures, tables, and data files are organized and referenced to ensure clarity, reproducibility, and ease of inspection by the reader.

Footnote

This mini-project is based entirely on the publicly available RNA-seq dataset and experimental framework described in **High-resolution mapping of transcriptional dynamics across tissue development reveals a stable mRNA–tRNA interface**, *and all analyses presented here are restricted to and derived from this study.*